AU tests of the monophyly constraints (Table S11)

p = 0.05

	ΔlnL	p-AU
sus-like (OCD-fold family)		
0. unconstrained maximum-likelihood tree (reference)	0	1.000
1. fungal sus-like + T-DNA sus forced monophyletic	14.829	0.264
2. fungal sus-like + alternative bacterial OCD-fold proteins forced monophyletic	176.12	7.0×10^{-50}
sus-like (OCD-fold family)		
0. unconstrained maximum-likelihood tree (reference; separate test set with the native fungal OCD-fold constraints for the sus-like genes; constraints 1-2 above re-tested here gave p = 0.26 and 5.9e-06)	0	1.000
3. fungal sus-like (52) + the native fungal OCD-fold homologs of the tree (2 native Sordariomycetes proteins and 4 fungal OCD-fold genes outside the sus-like clade) forced monophyletic	752.87	9.5×10^{-8}
4. fungal sus-like + the native fungal OCD-fold homologs + the 5 fungal OCD-fold proteins of the outgroup (11 fungal sequences) forced monophyletic	709.84	9.9×10^{-67}
cus-like (cucumopine synthase family)		
0. unconstrained maximum-likelihood tree (reference)	2.5899	0.528
1. fungal cus-like + T-DNA cus forced monophyletic	16.199	0.315
2. fungal cus-like + T-DNA cus + 4 bacterial references from the clade forced monophyletic	0	1.000
3. fungal cus-like + actinobacterial-type forced monophyletic	405.81	1.8×10^{-13}
4. fungal cus-like + mis group forced monophyletic	920.68	3.4×10^{-47}
5. fungal cus-like + other bacterial homologs forced monophyletic	1107.2	4.9×10^{-4}
6. all fungal cus-like in one clade (single acquisition)	129.43	6.4×10^{-5}
7. Boletales cus-like + T-DNA cus forced monophyletic, separate from other fungi	519.3	1.3×10^{-3}
8. Boletales cus-like + actinobacterial-type forced monophyletic	643.65	5.0×10^{-59}
9. Boletales cus-like + mis group forced monophyletic	1162.4	3.3×10^{-77}
10. Boletales cus-like + remaining fungal cus-like (references excluded) forced monophyletic	134.77	1.6×10^{-5}
cus-like outside Boletales (cucumopine synthase family; separate AU run)		
0. unconstrained maximum-likelihood tree (reference; separate test set for cus-like outside Boletales)	0	1.000
1. cus-like outside Boletales + T-DNA cus forced monophyletic, separate from Boletales cus-like	457.66	1.2×10^{-6}
2. cus-like outside Boletales + T-DNA cus + 4 bacterial references from the clade forced monophyletic, separate from Boletales cus-like	372.52	2.3×10^{-76}
3. cus-like outside Boletales + actinobacterial-type forced monophyletic	709.43	3.7×10^{-53}
4. cus-like outside Boletales + mis group forced monophyletic	1265	3.4×10^{-59}
5. cus-like outside Boletales + other bacterial homologs forced monophyletic	1414	1.1×10^{-41}
6. all cus-like outside Boletales in one clade (single acquisition)	454.89	9.0×10^{-67}
cus-like (cucumopine synthase family)		
0. unconstrained maximum-likelihood tree (reference; separate test set for the monophyly of the Boletales cus-like genes; the four Boletales constraints above re-tested here gave p = 5.1e-07, 1.3e-56, 1.3e-46 and 3.2e-05)	0	1.000
5. all 26 Boletales cus-like sequences in one clade (single acquisition by Boletales)	341.15	4.9×10^{-6}
GDPD family (acc-type GDPD and pKerr14a-type acs-like)		
0. unconstrained maximum-likelihood tree (reference)	0.34854	0.579
1. pKerr14a-type acs-like + T-DNA acs forced monophyletic	1723.7	1.4×10^{-60}
2. pKerr14a-type acs-like + pKerr14a acs references + Parasponia PON40589 forced monophyletic	17.528	0.415
3. pKerr14a-type acs-like + acc-type GDPD forced monophyletic	62.668	0.132
4. pKerr14a-type acs-like + native fungal GDPD forced monophyletic	3267	2.4×10^{-6}
5. acc-type GDPD + acc-locus references forced monophyletic	0	1.000
6. acc-type GDPD + T-DNA acs forced monophyletic	2121	5.9×10^{-47}
GDPD family (acc-type GDPD and pKerr14a-type acs-like)		
0. unconstrained maximum-likelihood tree (reference; separate test set with the native fungal GDPD constraint for the acc-type genes; constraints 1-6 above re-tested here gave p = 1.4e-31, 0.44, 0.16, 4.7e-61, 0.56 and 7.4e-11)	0	1.000
7. acc-type GDPD + native fungal GDPD forced monophyletic (native origin of the acc-type genes)	3497.6	1.1×10^{-95}
chs-like (chrysopine synthase; M24 aminopeptidase family)		
0. unconstrained maximum-likelihood tree (reference)	0	1.000
1. fungal chs-like + T-DNA chs seed forced monophyletic, without the other Ti-plasmid chs genes	133.26	2.0×10^{-48}
2. fungal chs-like + the four other Ti-plasmid chs genes forced monophyletic, without the seed	112.67	1.5×10^{-72}
3. fungal chs-like + ags seed forced monophyletic	314.43	8.8×10^{-4}
4. fungal chs-like + native fungal ags homologs forced monophyletic	49.57	1.7×10^{-3}
5. fungal chs-like + other bacterial M24 aminopeptidases forced monophyletic	320.27	4.1×10^{-15}
6. all fungal chs-like (Fusarium + Dactylonectria) in one clade (single acquisition)	4.5535	0.167
chs-like (chrysopine synthase; M24 aminopeptidase family)		
0. unconstrained maximum-likelihood tree (reference; separate test set: are the fungal chs-like genes nested within the Ti-plasmid chs clade?)	0	1.000
1. all 11 fungal chs-like (Fusarium + Dactylonectria) + the chs seed (pTi chrysopine synthase A0A2Z2PGA9) forced monophyletic (fungal genes nested within the Ti-plasmid chs clade next to seed)	133.26	1.9×10^{-4}
2. the 10 Fusarium chs-like + the chs seed (pTi chrysopine synthase A0A2Z2PGA9) forced monophyletic (Dactylonectria unconstrained)	147.28	8.4×10^{-62}
3. all 11 fungal chs-like (Fusarium + Dactylonectria) + Ti-plasmid chs WP_139103025.1 forced monophyletic (fungal genes nested within the Ti-plasmid chs clade next to b139)	133.26	1.9×10^{-4}
4. the 10 Fusarium chs-like + Ti-plasmid chs WP_139103025.1 forced monophyletic (Dactylonectria unconstrained)	147.28	4.3×10^{-60}
5. all 11 fungal chs-like (Fusarium + Dactylonectria) + Ti-plasmid chs WP_272871282.1 (A. tumefaciens) forced monophyletic (fungal genes nested within the Ti-plasmid chs clade next to b272871)	104.2	1.2×10^{-4}
6. the 10 Fusarium chs-like + Ti-plasmid chs WP_272871282.1 (A. tumefaciens) forced monophyletic (Dactylonectria unconstrained)	120.04	5.2×10^{-58}
7. all 11 fungal chs-like (Fusarium + Dactylonectria) + Ti-plasmid chs WP_272951408.1 (A. vitis) forced monophyletic (fungal genes nested within the Ti-plasmid chs clade next to b272951)	115.71	2.3×10^{-5}
8. the 10 Fusarium chs-like + Ti-plasmid chs WP_272951408.1 (A. vitis) forced monophyletic (Dactylonectria unconstrained)	133.79	2.6×10^{-40}
9. all 11 fungal chs-like (Fusarium + Dactylonectria) + Ti-plasmid chs WP_084631803.1 (A. larrymoorei) forced monophyletic (fungal genes nested within the Ti-plasmid chs clade next to b084)	75.791	3.8×10^{-6}
10. the 10 Fusarium chs-like + Ti-plasmid chs WP_084631803.1 (A. larrymoorei) forced monophyletic (Dactylonectria unconstrained)	94.92	3.7×10^{-5}
nos-subfamily opine dehydrogenase (Boletales; Rhypophila homolog, not assigned)		
0. unconstrained maximum-likelihood tree (reference)	0	1.000
1. Boletales nos-subfamily genes + T-DNA nos forced monophyletic (excluding non-T-DNA rhizobial paralogs)	35.625	3.0×10^{-3}
2. Rhypophila nos-subfamily genes + T-DNA nos forced monophyletic (excluding non-T-DNA rhizobial paralogs)	41.843	8.9×10^{-3}
3. all fungal nos-subfamily genes (Boletales + Rhypophila) in one clade (single acquisition)	4.9949	0.345
nos-subfamily opine dehydrogenase (Boletales; Rhypophila homolog, not assigned)		
0. unconstrained maximum-likelihood tree (reference; separate test set with the native fungal opine dehydrogenase constraints; constraints 1-3 of the first set re-tested here gave p = 0.0034, 0.0091 and 0.32)	0	1.000
4. Boletales nos-subfamily genes + native fungal opine dehydrogenases (19 sequences) forced monophyletic (native origin of the Boletales genes)	190.22	3.4×10^{-5}
5. Boletales + Rhypophila nos-subfamily genes + native fungal opine dehydrogenases forced monophyletic (single native origin of all fungal genes of the subfamily)	218.13	1.0×10^{-26}
nos-subfamily opine dehydrogenase (Boletales; Rhypophila homolog, not assigned)		
0. unconstrained maximum-likelihood tree (reference; separate test set for the Rhypophila genes and the native fungal opine dehydrogenases; constraints 1-5 of the first two sets re-tested here gave p = 0.0030, 0.0123, 0.34, 3.8e-44 and 1.2e-07)	0	1.000
6. Rhypophila genes + the two native opine dehydrogenases of Sordariomycetes adjacent to the subfamily (Cryphonectria parasitica, Gnomoniopsis smithogilvyi) forced monophyletic	51.33	2.0×10^{-3}
7. Rhypophila genes + all native opine dehydrogenases of Sordariomycetes (12 sequences) forced monophyletic	381.16	1.1×10^{-37}
8. Rhypophila genes + all native fungal opine dehydrogenases (19 sequences) forced monophyletic	154.25	2.1×10^{-46}
odh island (Fusarium; opine dehydrogenase family)		
0. unconstrained maximum-likelihood tree (reference)	0.00035445	1.000
1. odh-island ODH + all native fungal ODH forced monophyletic	305.01	3.5×10^{-6}
2. odh-island ODH + native Sordariomycetes ODH (Apiospora, Astrocystis) forced monophyletic (vertical-inheritance expectation)	466.77	4.0×10^{-4}
3. control: native fungal ODH forced monophyletic (odh-island ODH unconstrained)	36.536	6.4×10^{-3}
4. odh-island ODH inside the ML T-DNA clade (ocs/nos/vis seeds + Rhypophila/Butyriboletus + A. larrymoorei WP_309770762) forced monophyletic	125.01	2.5×10^{-58}
5. odh-island ODH + T-DNA ocs/nos/vis seeds only forced monophyletic	266.99	3.5×10^{-57}
6. control: odh-island ODH + A. larrymoorei WP_309770759 forced monophyletic (clade of the ML tree)	0	1.000
7. odh-island ODH + all other bacterial ODH (both A. larrymoorei ODH excluded) forced monophyletic	283.36	2.1×10^{-7}
8. odh-island ODH inside the nos subfamily (nos seed P00386 + Gillisella) forced monophyletic	194.05	1.4×10^{-51}

- unconstrained ML tree (reference)
- constraint not rejected (p ≥ 0.05)
- constraint rejected (p < 0.05)

10^{0} 10^{-20} 10^{-40} 10^{-60} 10^{-80}

p-AU (log scale)